

B. TECH
(SEM V) THEORY EXAMINATION 2022-23
BIOFUEL & ALCOHOL TECHNOLOGY

Time: 3 Hours**Total Marks: 100****Note:** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief. 2 x 10 = 20

- (a) What are the fermentation conditions for the ethanol production?
- (b) What do you understand by single cell protein?
- (c) What do you understand by fed batch fermentation process?
- (d) What is fivefold purpose of cooking?
- (e) Enlist various feedstocks used in alcohol industries.
- (f) What is the use of denatured spirit?
- (g) Describe reduction and blending of spirits.
- (h) What do you understand by biomass?
- (i) What do you understand by thermal gasification of biomass?
- (j) Why do we use bioenergy?

SECTION B

2. Attempt any three of the following: 10 x3= 30

- (a) Enlist the types and properties of raw materials used in alcohol industries.
- (b) Differentiate between batch and continuous fermentation techniques.
- (c) Write short note on-
 - (i) Biochemistry of alcoholic fermentation
 - (ii) Fusel oil separation from alcohol
- (d) Compare the direct, thermochemical and biochemical biomass conversion processes.
- (e) What are the different types of biomass fuel? Explain.

SECTION C

3. Attempt any one part of the following: 10x1 = 10

- (a) What are the different pre-processing methods used for raw materials? What are the advantages of pre-processing? How the raw materials are stored? Discuss with examples.
- (b) What is the role of yeast in alcohol production? Enlist the different yeast strains used in alcohol (respective) industries.

4. Attempt any one part of the following: 10x1 = 10

- (a) Describe various cooking systems used in alcohol industries.
- (b) Write short note on-
 - (i) Bio-still Fermentation
 - (ii) Wet milling of grains

5. Attempt any *one* part of the following: 10x1 = 10

- (a) What do you understand by cellulosic feed stocks? What characteristics of cellulosic feed stocks make them fit for alcohol fermentation? How it is used in alcohol production?
- (b) Explain the process of distillation.

6. Attempt any *one* part of the following: 10x1 = 10

- (a) Write short note on-
 - (i) Parameters affecting alcoholic fermentation
 - (ii) Scale-up process
- (b) What are the byproducts of alcohol industries? How it is used? Explain.

7. Attempt any *one* part of the following: 10x1 = 10

- (a) What is anaerobic digestion? What are its stages? Write a note on quantitative evaluations of the anaerobic digestion process.
- (b) Briefly explain the processes used to convert biomass into biofuels by industries.

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